BlackboxNLP 2026

Duo Project Plan & Work Division

The Myth of Universal Concept Directions — Under RLHF

Paper Track	Archival Full Paper — Track 1 (Original Research)
Workshop	BlackboxNLP @ EMNLP 2026, Budapest, Hungary — Oct 29, 2026
Team Size	2 people (Person A + Person B)
Compute	Kaggle ~30h/week free GPU budget
Fallback	#6 — Logit Lens replication (same pipeline, lower risk)

📅 Deadline Overview

All deadlines are 11:59 PM UTC-12:00 (Anywhere on Earth). Internal targets are set 2–3 days before each official deadline.

Official Deadline	What's Due	Our Internal Target	Risk Level
June 30, 2026	Expression of Interest — Reproducibility Challenge	June 27	🟡 Optional but recommended
July 17, 2026	Direct paper submission via OpenReview	July 14	🔴 HARD — primary deadline
July 30, 2026	Reproducibility Challenge submission	July 27	🟡 Fallback track option
Aug 28, 2026	ARR pre-reviewed paper commitment	Aug 25	🟢 Only if ARR route taken
Sep 20, 2026	Camera-ready version due	Sep 17	🟢 Post-acceptance only

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Phase 1 — Foundations & Setup

🗓️ **NOW → June 27 (Internal Target: June 27) | Expression of Interest for Reproducibility Challenge**

This phase sets up everything needed before any experiments run. Both people work in parallel. Goal: by June 27, submit the Expression of Interest form AND have a working pilot on Gemma 2 2B.

Person A — Research & Framing

- Draft the Expression of Interest (2–3 paragraphs): research question, method, model family, expected contribution
- Write a 1-page research proposal outline: title, abstract draft, 3 key hypotheses
- Identify and download datasets: TruthfulQA (honesty pairs), AdvBench (refusal pairs)
- Literature review: Zou et al. 2023 (Representation Engineering), Hernandez et al. (concept vectors), Marks & Tegmark (geometry of truth)
- Define the 4 domains per concept and begin writing contrastive prompt pairs (target: 80 pairs by end of Phase 1)

Person B — Infrastructure & Compute

- Set up Kaggle notebooks: install TransformerLens, HuggingFace Transformers, bitsandbytes
- Load Gemma 2 2B (Base + Instruct) and verify forward pass works end-to-end
- Write activation extraction script: hook residual stream at every layer, cache to disk efficiently
- Test pipeline on 10 dummy prompt pairs — verify shapes, caching, and reproducibility
- Set up shared GitHub repo: notebooks, data folders, results folder structure

Joint Tasks

- Agree on concept definitions: what counts as 'refusal' vs 'truthfulness' — write a 1-paragraph operational definition for each
- Review and finalize model family choice: Gemma 2 2B for pilot, Qwen 2.5 3B for full study
- Submit Expression of Interest by June 27

Task	Person A	Person B	Status
Expression of Interest draft	Write + lead	Review + edit	Both
Literature review (4 key papers)	Lead	Support	A
Dataset download + format	TruthfulQA	AdvBench	Both
Prompt pair writing (80 pairs)	Lead	Support	A
Kaggle environment setup	Support	Lead	B
Activation extraction script	Review	Write	B
GitHub repo structure	Support	Lead	B
Pilot run (Gemma 2 2B, 10 pairs)	Observe + log	Run	B
Concept operational definitions	Draft	Review	Both

Phase 2 — Data + Full Extraction

🗓️ June 28 → July 10 (Buffer for July 14 Internal Target) | Full Experiment Pipeline Ready

This is the heaviest experimental phase. All prompt pairs finalized, all activations extracted across both concepts and all domains for Qwen 2.5 3B Instruct. By July 10, you should have results in hand and be writing.

Person A — Prompt Engineering & Data

- Complete all contrastive prompt pairs: 150–250 per concept × 2 concepts = 300–500 total pairs
  - Refusal domains: violence, illegal activity, medical/legal advice, privacy
  - Honesty domains: factual trivia, math, politics/opinion, personal advice
- Add harmlessness as a third concept (pairs across 3–4 domains) — strengthens RLHF comparison story
- Write domain-split annotation guide so both team members label consistently
- Validate prompt pairs: ensure each pair is minimal-contrast (only the sensitive dimension changes)
- Pull pairs from TruthfulQA and AdvBench to anchor to prior work — document which pairs are sourced vs. hand-written

Person B — Full Extraction Runs

- Run full extraction on Qwen 2.5 3B Instruct: both concepts (refusal + honesty), all 4 domains each, all layers
- Cache activations to disk with clear naming conventions (model_concept_domain_layer)
- Compute global direction per concept: difference-of-means across ALL domains
- Compute per-domain directions: difference-of-means separately for each of the 4 domains
- Run extraction on Qwen 2.5 3B Base checkpoint (RLHF comparison axis)
- If harmlessness pairs ready from Person A: add third concept extraction

Joint Tasks

- Midpoint check (around July 3–4): review extraction results together, sanity check direction vectors
- Decide if null results are emerging early enough to pivot to fallback #6 (Logit Lens)

Task	Person A	Person B	Status
Prompt pairs — Refusal (150–250)	Write	Review	A
Prompt pairs — Honesty (150–250)	Write	Review	A
Prompt pairs — Harmlessness (100+)	Write	Review	A
Domain annotation guide	Lead	Support	A
Full extraction — Qwen 3B Instruct	Monitor	Run	B
Full extraction — Qwen 3B Base	Monitor	Run	B
Global direction computation	Verify	Implement	B
Per-domain direction computation	Verify	Implement	B
Midpoint sanity check (July 3–4)	Both	Both	Both

Pivot decision if null result	Both	Both	Both

Phase 3 — Core Analysis & Writing

🗓️ **July 10 → July 14 (INTERNAL HARD TARGET — 3 days before July 17 official) | Full paper draft submitted to OpenReview**

This is the crunch phase. Experiments run, paper written, submitted. Person A owns writing, Person B owns final experiments and figures. No new experiments after July 12.

Person A — Writing

- Write Introduction: motivation, why concept directions matter, RLHF safety stakes (1 strong paragraph on real-world implications)
- Write Related Work: Zou et al., Hernandez et al., Marks & Tegmark, plus 2–3 activation steering papers
- Write Methods section: contrastive prompt construction, activation extraction, direction computation, steering protocol
- Write Results section as soon as Person B delivers figures
- Write Discussion: what the piecewise/null result means for the activation steering literature
- Write Conclusion + Limitations
- Handle OpenReview submission logistics: format in ACL LaTeX template, check page limits (8 pages + refs)

Person B — Final Experiments & Figures

- Angular dispersion analysis: cosine similarity between per-domain vs global directions — produce heatmap figure
- Steering transfer test: apply global direction to held-out prompts per domain, measure refusal/truthfulness rate shift
- Cross-domain transfer: apply domain A direction to domain B prompts — produce transfer matrix figure
- Bootstrap confidence intervals on all effect sizes (min 1000 bootstrap samples)
- RLHF comparison: compute Base vs Instruct gap in angular dispersion — produce comparison figure
- PCA/UMAP visualization of per-domain direction clusters — key geometric figure for the paper
- Deliver all figures (publication quality, 300 DPI) to Person A by July 11

Joint Tasks — Review Sprint (July 12–14)

- Both read full draft: check consistency between claims and figures
- One full proofreading pass each
- Cross-check all citations against actual papers
- Final OpenReview submission — both approve before submitting

Task	Person A	Person B	Status

Introduction + motivation	Write	Review	A
Related Work section	Write	Review	A
Methods section	Write	Review	A
Angular dispersion + heatmap	Review	Implement	B
Steering transfer test + figure	Review	Implement	B
Cross-domain transfer matrix	Review	Implement	B
Bootstrap CIs on all effects	Verify	Implement	B
RLHF comparison figure	Review	Implement	B
PCA/UMAP geometry figure	Review	Implement	B
Results + Discussion writing	Write	Data support	A
Full draft review (July 12–13)	Both	Both	Both
OpenReview submission (July 14)	Lead logistics	Support	Both

Phase 4 — Reproducibility Track (Optional)

> 🗓️ **July 15 → July 27 (Internal Target — 3 days before July 30 official) | Reproducibility Challenge submission**

Only pursue this if: (a) main paper submitted, AND (b) your results show interesting failure modes worth documenting separately. This is the fallback #6 (Logit Lens replication) or a reproducibility extension of your own findings.

If Pursuing Fallback #6 — Logit Lens

- Person B: run Logit Lens analysis on same Qwen 2.5 3B checkpoints using the already-cached activations
- Person A: write a short 4-page paper documenting replication across architectures (Gemma vs Qwen)
- Joint: submit by July 27

If Extending Main Paper

- Document the full experimental setup in a reproducibility appendix
- Release code + prompt pairs on GitHub with a clear README
- Submit the reproducibility report by July 27

Phase 5 — Camera Ready (If Accepted)

🗓️ **Aug 25 (Internal Target — 3 days before Aug 28)** |	**ARR commitment (if ARR route)**
🗓️ **Sep 17 (Internal Target — 3 days before Sep 20)** |	**Camera-ready version**

Only relevant post-acceptance (Sep 8 notification). Camera-ready typically means: incorporating reviewer feedback, final figures, author list finalization, and ACL Anthology metadata.

- Address all reviewer comments — Person A leads, Person B handles any additional experiments requested
- Final figure polish (font sizes, labels, colorblind-safe palette)
- Write an acknowledgments section
- Submit camera-ready by Sep 17

⚡ Compute Budget Allocation

Session	Task	Who Runs	Est. GPU hrs	Phase
1	Pilot — Gemma 2 2B, debug full pipeline, 10 pairs	Person B	~5h	Phase 1
2	Full extraction — Qwen 2.5 3B Instruct, both concepts, all domains	Person B	~6h	Phase 2
3	Direction computation + angular dispersion analysis (CPU-heavy)	Person B	~3h	Phase 3
4	Steering + cross-domain transfer experiments	Person B	~6h	Phase 3
5	Repeat extraction on Base checkpoint (RLHF comparison)	Person B	~6h	Phase 2–3
6	Buffer: re-runs, extra seeds, OR Logit Lens fallback (#6)	Person B	~4–6h	Flex
TOTAL	Fits within 1–2 weeks Kaggle quota with buffer	—	~26–32h	—

⚠️ Risk Register & Contingency Plans

Risk	Likelihood	Contingency
Null result — global direction is fine everywhere	Medium	Frame as a strong positive reproducibility result; submit to Reproducibility Track instead. Pair with Logit Lens fallback.

Kaggle GPU quota exhausted mid-experiment	Medium	Switch to Gemma 2 2B for all remaining runs (faster). Use 4-bit quantization for 3B models. Cache aggressively to avoid re-running.
Prompt pairs lack quality / domain coverage	Low-Med	Pull more pairs from TruthfulQA and AdvBench. Person B can review and filter pairs while Person A writes new ones.
TransformerLens doesn't support chosen model well	Low	Fall back to manual hook registration via HuggingFace. Llama 3.2 3B has best TransformerLens community support as backup.
Team bandwidth mismatch / one person unavailable	Medium	Person A can run extractions if needed (scripts are documented). Scope down to 2 concepts instead of 3 if deadline is tight.
Paper rejected / desk-rejected for scope	Low	BlackboxNLP is a workshop — acceptance bar is lower than main EMNLP. Strong negative results explicitly welcome per CFP.

🛠️ Tooling & Infrastructure

- **Activation caching, hooks, activation addition (steering)** TransformerLens
- **Model loading, 4-bit/8-bit quantization for 3B models** HuggingFace Transformers + bitsandbytes
- **PCA, UMAP, cosine similarity, bootstrap confidence intervals** scikit-learn / numpy / scipy
- **Honesty/truthfulness contrastive pairs — anchor to prior work** TruthfulQA
- **Refusal benchmark pairs — used directly as prompt set source** AdvBench
- **Score refusal rate + coherence of steered outputs (use Instruct model itself)** A small judge model
- **Heatmaps, transfer matrices, PCA scatter plots — all publication quality** Matplotlib / seaborn
- **Paper formatting — download from ACL Anthology GitHub** ACL LaTeX template
- **Submission portal — both team members should create accounts now** OpenReview account
- **Code, notebooks, prompt pairs, results — set up in Phase 1** Shared GitHub repo
- **~30h/week GPU budget — Person B primary account; Person A as backup** Kaggle (free tier)

📚 Must-Read Papers Before Writing

- **The canonical concept-direction paper. Your work is a stress-test of their claims — cite heavily.** Zou et al. (2023) — Representation Engineering
- **Establishes that truth has linear structure in LLM activations. Direct prior work.** Marks & Tegmark (2023) — The Geometry of Truth
- **Supports linear representation hypothesis. Good for related work section.** Hernandez et al. (2023) — Linearity of Relation Decoding
- **Caution about universality of findings — supports your framing.** Lieberum et al. (2023) — Does Circuit Analysis Interpretability Scale?
- **RLHF alignment fragility literature — directly motivates your RLHF angle.** Wei et al. (2024) — Assessing the Brittleness of Safety Alignment

- **Dataset paper — cite when describing your honesty prompt set.** TruthfulQA (Lin et al., 2022)

📝 Final Notes

This is a strong project. The compute is realistic, the question is genuinely contested, and the RLHF comparison axis is novel. A few things to keep front of mind:

- Don't presuppose your result in the title — keep it as a question until you have data
- The special reproducibility track (July 30 deadline) is your safety net — keep the Logit Lens fallback warm
- Cite Zou et al. prominently and frame your work as a rigorous boundary-testing study, not a refutation
- The geometric figure (PCA/UMAP of per-domain directions) will be the most-cited figure — invest time in making it beautiful
- Person A should start the ACL LaTeX template immediately — formatting takes longer than expected
- Add harmlessness as a third concept only if Person A can deliver pairs before July 3 — don't let it slip the timeline

Generated for BlackboxNLP 2026 — Good luck!